

THE HILBERT SERIES OF THE INVARIANTS OF TERNARY SEPTICS, OCTICS AND NONICS

ANONYMOUS

ABSTRACT. We determine the Hilbert series of the rings of $\mathrm{SL}_3(\mathbb{C})$ -invariants of ternary forms of degrees 7, 8 and 9. Within the literature we found, the largest degree previously completed was 6 (Bedratyuk–Xin, 2011). The septic series equals the candidate $R_7(t)$ of version 0.1.0 of this work, where it was proved only through degree 760 and conditionally beyond. The least denominator of R_7 has degree 1386. We also determine the octic series, with a least denominator of degree 2331, and the nonic series ($d = 9$), with a least denominator of degree 1393. The numerators are palindromic of degrees 2286 and 1338.

The method is not new. Derksen’s universal denominators bound the pole orders of the Hilbert series at roots of unity by fixed-point dimensions, computable from the torus weights. The functional equation fixes the degree of the numerator (Knop; Stanley). Finitely many exactly computed coefficients then determine the series, as in Makam’s work on matrix semi-invariants. Our contribution is the application to ternary forms. We compute the torus bounds for $d = 7, 8, 9$ by an exhaustive enumeration, which a separate program reproduces blind. We also compute the required coefficients exactly: for septics, the degree-760 data already suffice. The same pipeline re-derives the known series for $d = 5, 6$ exactly.

This version also corrects two attribution omissions in version 0.1.0.

1. INTRODUCTION

Let $I_{3,d} = \mathbb{C}[S^d\mathbb{C}^3]^{\mathrm{SL}_3}$ be the ring of invariants of ternary forms of degree d , graded by degree, and let $H_d(t) = \sum_n \dim(I_{3,d})_n t^n$. The series is classical for $d \leq 4$ (see [2]). Bedratyuk and Xin [2] computed $d = 5, 6$ by MacMahon partition analysis. For $d = 7$, Bedratyuk [1] gave the coefficients through degree 21. We found no complete series for any $d \geq 7$ (search record in `PRIOR_ART.md`; MathSciNet and zbMATH were not searched).

Version 0.1.0 of this work [12] gave a rational function R_7 and proved $H_7 = R_7$ through degree 760. Full equality was proved only under a hypothesis on a homogeneous system of parameters, whose existence was unknown.

Theorem 1 (plane septics; computer-assisted). $H_7(t) = R_7(t)$. *The least denominator of H_7 is $\prod_r \Phi_r^{m_r}$, a product of cyclotomic polynomials of degree 1386 with the multiplicities of Table 1. The numerator in lowest terms is palindromic of degree 1350.*

Theorem 2 (octics and nonics; computer-assisted). $H_8(t) = N_8(t)/D_8(t)$, where $D_8 = \prod_r \Phi_r^{m_r}$ has degree 2331 with the multiplicities of Table 2, and N_8 is palindromic of degree 2286. Likewise $H_9(t) = N_9(t)/D_9(t)$, where D_9 has degree 1393 (Table 2) and N_9 is palindromic of degree 1338. The polynomials are in `data/octic/d8_rational.json` and `data/nonic/d9_rational.json`. The first coefficients are

$$H_8 = 1 + t^3 + 6t^6 + 81t^9 + 1990t^{12} + 46543t^{15} + 873667t^{18} + \dots,$$

$$H_9 = 1 + t^4 + 6t^6 + 3t^7 + 46t^8 + 118t^9 + 612t^{10} + 2012t^{11} + 8300t^{12} + \dots.$$

Date: Version 0.2.0-candidate, 26 September 2026. Unrefereed candidate. Follow-up to version 0.1.0-candidate, doi:10.5281/zenodo.22974308.

Method and what is new. The route to both theorems has four steps, and each is known.

- (1) *Pole orders from fixed points.* Derksen [4] defines the universal denominator $\text{udenom}(M, t)$ of a graded module M , the least common multiple of the denominators of the Hilbert series of all submodules. He proves that $\text{udenom}(M, t) = \prod_r \Phi_r^{m_r}$ with $m_r = \dim M/I^{[r]}M$, where $I^{[r]}$ is generated by the homogeneous elements whose degree is not divisible by r [4, Thm. 1.10].
- (2) *Reduction to the torus.* For G connected reductive and T a maximal torus, $\text{udenom}(S^G)$ divides $\text{udenom}(S^T)$ [4, Thm. 3.4]. The torus version is explicit: it is a maximum of $\#I - \text{rank } \Omega_I$ over weight subsets I satisfying a lattice condition [4, Thm. 5.1].
- (3) *Functional equation.* The rings are Gorenstein with $\deg H_d = -\dim V$ (Knop’s criterion, as used in [12]). So $H_d \cdot B$ is a palindromic polynomial of known degree for every denominator B .
- (4) *Finite determination.* Half the coefficients of that polynomial then determine H_d . Makam used exactly this combination of universal denominators, Knop’s theorem and the functional equation for matrix semi-invariants [7, §§3,6].

Our contributions are:

- the exhaustive computation of the torus bounds for $S^d\mathbb{C}^3$ with $d = 7, 8, 9$, reproduced by an independently written program;
- the observation that for $d = 7$ these bounds are within 2 of the true multiplicities at every order, so that the exact data of [12] already suffice;
- the exact coefficients for $d = 8, 9$;
- the certificates.

Corrections to version 0.1.0. Version 0.1.0 [12] is immutable. We record two attribution omissions here. Neither affects any stated result.

- (1) Its §7 describes an a priori denominator for H_7 , obtained from extreme rays of the weight cone, as available “by standard means”, without citing Derksen [4], which gives a sharper version.
- (2) It presents the combination of a denominator theorem, the functional equation and finitely many exact coefficients as part of its methodological contribution. The same strategy appears in Makam [7, §6], which [12] cites only for Knop’s criterion. The contribution of [12] is the new cases, the implementation and the certificates.

2. THE CERTIFICATE

Let $G = \text{SL}_3$, $V = S^d\mathbb{C}^3$, $A = \mathbb{C}[V]^G$ and $\delta = \dim V - 8$, the Krull dimension. Write $H = H_d = \sum a_n t^n$.

Proposition 3. *Let $T \subset G$ be the diagonal torus. Write $\text{udenom}(\mathbb{C}[V]^T, t) = \prod_r \Phi_r^{b_r}$, and let $B = \prod_r \Phi_r^{p_r}$ with $p_r = \min(\delta, b_r)$. Then $P = H \cdot B$ is a polynomial with $P(t) = \varepsilon t^K P(1/t)$, where $K = \deg B - \dim V$ and $\varepsilon = (-1)^\delta \cdot (-1)^{p_1}$. Consequently H is determined by $a_0, \dots, a_{\lfloor K/2 \rfloor}$.*

Proof. By [4, Thm. 3.4], $\text{udenom}(A)$ divides $\text{udenom}(\mathbb{C}[V]^T)$. The pole order of H at a primitive r -th root of unity is at most the exponent of Φ_r in $\text{udenom}(A)$, since A is a submodule of itself. So it is at most b_r . It is also at most δ : by homogeneous Noether normalisation, A is finite over a polynomial ring in δ variables. Hence $H \cdot B$ has no finite poles and is a polynomial.

The functional equation $H(1/t) = (-1)^{\delta \dim V} H(t)$ (§2.1) and $B(1/t) = (-1)^{p_1} t^{-\deg B} B(t)$ give the symmetry. A polynomial of degree at most K with this symmetry is determined by its coefficients of degree at most $\lfloor K/2 \rfloor$, and these are those of $H \cdot B$. $\square$

2.1. **The functional equation.** The argument of [12, §3.1] applies to every $d \geq 5$.

- A is Cohen–Macaulay [6]. It is factorial because SL_3 has no nontrivial characters, and hence Gorenstein [9].
- $S^d \mathbb{C}^3$ is not coregular for $d \geq 4$ [10], so it is 2-large [5, Thm. 3.7].
- Knop’s criterion, as stated in [8, Thm. 3.3], then gives $\deg H = -\dim V$.

Stanley’s theorem [11] gives $H(1/t) = (-1)^{\delta} t^{\dim V} H(t)$. For $d = 7$: $\dim V = 36$ and $\delta = 28$. For $d = 8$: $\dim V = 45$ and $\delta = 37$. For $d = 9$: $\dim V = 55$ and $\delta = 47$.

3. THE TORUS BOUNDS

The torus weights of $S^d \mathbb{C}^3$ are $w = (a - c, b - c)$ for $a + b + c = d$. By [4, Thm. 5.1] and the proof of [4, Thm. 3.4], b_r is the largest dimension of a torus quotient $V_S // T$. Here $S = \{w : \chi_w(t) = \zeta\}$ is the set of weights on which some torus element t acts by a primitive r -th root of unity ζ , and

$$\dim V_S // T = |S_0| - \mathrm{rank} S_0, \quad S_0 = \{w \in S : -w \in \mathrm{cone}(S)\}.$$

We enumerate every such level set exactly. Write $S = W \cap \phi^{-1}(\zeta)$ for a homomorphism $\phi : \mathbb{Z}^2 \rightarrow \mathbb{C}^*$.

- If the differences of S span a rank-2 lattice, then $\ker \phi$ has cyclic quotient of order at most the largest $|\det|$ of two weight differences (147, 192 and 243 for $d = 7, 8, 9$). All such kernels are enumerated.
- If S lies on a line, then $\dim V_S // T = 0$ unless the line passes through 0. In that case S lies in a residue class of multiples of a primitive direction, and these are enumerated up to the diameter of the line.
- Singletons contribute 0.

The program is `scripts/pole_bounds.py`. A second program, written blind by a separate agent from the definition alone, enumerates the lattice condition of [4, Thm. 5.1] directly. It gives identical values of b_r for $d = 5, \dots, 9$, and for $d \leq 7$ it found an explicit torsion witness for every value. For the known cases $d = 5, 6$, the bounds are at least the true multiplicities of the least denominator everywhere, and positive exactly on its pole set.

TABLE 1. Plane septics: the bound $p_r = \min(28, b_r)$ and the multiplicity m_r of Φ_r in the least denominator of H_7 , as p_r/m_r . For all other r , $p_r = m_r = 0$.

1:28/28, 2:18/16, 3:28/28, 4:10/8, 5:9/7, 6:18/16, 7:3/3, 8:4/4, 9:13/11, 10:4/4, 11:2/2, 12:10/8,
13:2/2, 14:1/1, 15:9/7, 16:1/1, 17:1/1, 18:8/6, 19:1/1, 20:1/1, 21:3/3, 22:1/1, 23:1/1, 24:4/4, 25:1/1,
26:1/1, 27:3/3, 29:1/1, 30:4/4, 33:2/2, 36:3/3, 39:2/2, 42:1/1, 45:2/2, 48:1/1, 51:1/1, 54:2/2, 57–78
in steps of 3: 1/1 each, 87:1/1, 90:1/1, 108:1/1.

4. PLANE SEPTICS: PROOF OF THEOREM 1

For $d = 7$ the bound $B = \prod \Phi_r^{p_r}$ has degree 1452, so $K = 1416$ and $\varepsilon = +1$. By Proposition 3, H_7 is determined by $a_0, \dots, a_{708}$. These are known as exact integers for $n \leq 760$ [12]. They come from a weight-counting computation modulo 15 distinct 16-bit primes, which the package regenerates from scratch in about three minutes.

The script `certify_septic_unconditional.py` checks, in exact integer arithmetic, that:

- the least denominator D_7 of R_7 divides B ;
- $Q = R_7 B$ is a polynomial of degree at most 1416 with $Q(t) = t^{1416} Q(1/t)$;
- the series of R_7 equals the exact a_n for every $n \leq 708$.

The script regenerates the torus bounds itself. Hence H_7B and Q are reciprocal polynomials of degree 1416 that agree in their lower halves, so they are equal, and $H_7 = R_7$.

The bound exceeds the true multiplicity only at $r \in \{2, 4, 5, 6, 9, 12, 15, 18\}$, and by exactly 2 each time. The same pattern occurs for $d = 5, 6$. Presumably the bound becomes exact if centralisers larger than T are used in the proof of [4, Thm. 3.4]; we do not need this.

5. OCTICS AND NONICS: PROOF OF THEOREM 2

Here no candidate is needed: Proposition 3 produces the series directly from the exact coefficients.

Octics. For $d = 8$: $\dim V = 45$, $\delta = 37$, and the bound B has degree 2433 over 62 orders. So $K = 2388$, $\varepsilon = +1$, and $a_0, \dots, a_{1194}$ determine H_8 . We computed them by the grid method at six distinct 62-bit primes (372 bits). This exceeds twice $\binom{1194+44}{44} < 2^{271}$, so the Chinese remainder theorem recovers the integers. The computation took 89 minutes on an eight-core laptop. The script `certify_ternary.py` forms $P = H \cdot B$ from these coefficients, completes it by palindromy, and reduces P/B to lowest terms. It checks that the pole order at $t = 1$ equals $\delta = 37$ and that the series is nonnegative.

Nonics. For $d = 9$: $\dim V = 55$ and $\delta = 47$. The bound has degree 1427, so $a_0, \dots, a_{686}$ suffice. We computed them at six 62-bit primes, about five minutes each.

Three checks lie outside the determination itself (`check_ternary_extra.py`):

- the determined series predict the coefficients modulo a fresh 31-bit prime up to degree 1600 for $d = 8$ and 1000 for $d = 9$, far beyond the data used;
- they agree with an independent weight-counting program, which uses no roots of unity, modulo two 16-bit primes up to degree 450 for $d = 8$ and 400 for $d = 9$;
- the same pipeline, applied to $d = 5$ and $d = 6$, re-derives Bedratyuk and Xin’s rational functions exactly.

As for $d = 5, 6, 7$, the torus bound exceeds the true multiplicity only at a few small orders, by exactly 2: at $r \in \{2, 4, 5, 6, 7, 9, 12, 15, 18, 21\}$ for $d = 8$, and at $r \in \{2, 4, 5, 7, 8\}$ for $d = 9$.

TABLE 2. Multiplicities m_r of Φ_r in the least denominators of H_8 and H_9 ($r:m_r$; all others are 0).

$d = 8$	1:37, 2:16, 3:37, 4:8, 5:9, 6:16, 7:7, 8:4, 9:14, 10:4, 11:2, 12:8, 13:2, 14:3, 15:9, 16:1, 17:2, 18:6, 19:1, 20:1, 21:7, 22:1, 23:1, 24:4, 25:1, 26:1, 27:4, 28:1, 29:1, 30:4, 31:1, 33:2, 35:1, 36:3, 37:1, 39:2, 41:1, 42:3, 45:3, 48:1, 49:1, 51:2, 54:2, 57:1, 60:1, 63:2, 66:1, 69:1, 72:1, 75:1, 78:1, 81:1, 84:1, 87:1, 90:1, 93:1, 105:1, 108:1, 111:1, 123:1, 126:1, 147:1
$d = 9$	1:47, 2:26, 3:16, 4:14, 5:11, 6:8, 7:9, 8:8, 9:4, 10:6, 11:3, 12:4, 13:3, 14:5, 15:3, 16:4, 17:2, 18:2, 19:2, 20:3, 21:2, 22:2, 23:2, 24:2, 25:2, 26:1, 27:1, 28:2, 29:1, 30:1, 31:1, 32:2, 33:1, 34:1, 35:1, 36:1, 37:1, 38:1, 40:1, 41:1, 42:1, 43:1, 44:1, 46:1, 47:1, 48:1, 49:1, 50:1, 55:1, 56:1, 64:1

6. VERIFICATION

Table 3 lists the checks. ‘Exact’ means integer equality; ‘mod p ’ means agreement modulo a prime, which is evidence, not proof.

7. LIMITATIONS

- The theorems are computer-assisted. They rest on the cited theorems, on the enumeration programs, and on the correctness of the coefficient computations: for septics, exact weight counting; for octics, exact grid coefficients modulo six 62-bit primes, checked out of sample and against independent weight counting.

TABLE 3. Checks. “Exact” means integer equality; “mod p ” means agreement modulo the stated prime(s).

Check	Result
Torus bounds b_r , $d = 5, 6, 7$, recomputed blind by a separate program	identical
Torus bounds versus the known least denominators, $d = 5, 6$ (Bedratyuk–Xin)	$b_r \geq m_r$ everywhere; same support
Septic certificate (Proposition 3) with exact a_n , $n \leq 708$	passes
Exact septic a_n , $n \leq 760$, regenerated from 15 weight-counting runs	exact
Torus bounds b_r , $d = 8, 9$, recomputed blind	identical
Method re-derives Bedratyuk–Xin’s H_5 and H_6 from fresh grid data	exact, equal
H_8 : exact a_n , $n \leq 1194$ (six 62-bit primes); determination; pole order 37	passes
H_8 : prediction modulo a 31-bit prime, $n \leq 1600$	mod p , agrees
H_8 : independent weight counting modulo 65521 and 65519, $n \leq 450$	mod p , agrees
H_9 : exact a_n , $n \leq 686$; determination; pole order 47	passes
H_9 : prediction modulo a 31-bit prime, $n \leq 1000$; weight counting, $n \leq 400$	mod p , agrees
Adversarial review of the septic argument by a model from another vendor (GPT-5.6), two rounds	round 1 invalid on write-up gaps; round 2 valid with minor gaps, repaired

- No step is formally verified.
- Knop’s and Formanek’s theorems are used through the restatements in [8, 3].
- All review is model-based.
- The next case, $d = 10$, needs exact coefficients up to degree 3249 with the torus bound, roughly 100 CPU-hours with the present engine. The excess of the torus bound over the truth, always 2 and at a few small orders, suggests that sharper bounds from larger centralisers would reduce this.

PROVENANCE AND AI-USE STATEMENT

This work was produced by an AI research agent (Claude, Anthropic) under the direction of the publisher, Evidence Press. The problems, computations, proofs and text are AI-generated, and no human mathematical contribution is claimed. Parts of the verification were written by separate AI agents working from the definitions alone. No human specialist has checked the work.

REFERENCES

- [1] L. Bedratyuk, Analogue of the Sylvester–Cayley formula for invariants of ternary form, *Ukrainian Math. J.* 62 (2010) 1810–1821; arXiv:0806.1920.
- [2] L. Bedratyuk, G. Xin, MacMahon partition analysis and the Poincaré series of the algebras of invariants of ternary and quaternary forms, *Linear Multilinear Algebra* 59 (2011) 789–799, doi:10.1080/03081087.2010.536763.
- [3] A. Berele, Denominators for one variable Poincaré series of generic matrices, *J. Algebra* 618 (2023) 195–213; arXiv:2208.06392.
- [4] H. Derksen, Universal denominators of Hilbert series, *J. Algebra* 285 (2005) 586–607, doi:10.1016/j.jalgebra.2004.10.029.
- [5] H.-C. Herbig, G. W. Schwarz, The Koszul complex of a moment map, arXiv:1205.4608.
- [6] M. Hochster, J. L. Roberts, Rings of invariants of reductive groups acting on regular rings are Cohen–Macaulay, *Adv. Math.* 13 (1974) 115–175.
- [7] V. Makam, Hilbert series and degree bounds for matrix (semi-)invariants, *J. Algebra* 454 (2016) 14–28, doi:10.1016/j.jalgebra.2016.01.032.
- [8] V. Makam, arXiv version of [7]: arXiv:1510.08420.

- [9] M. P. Murthy, A note on factorial rings, *Arch. Math.* 15 (1964) 418–420.
- [10] G. W. Schwarz, Representations of simple Lie groups with regular rings of invariants, *Invent. Math.* 49 (1978) 167–191.
- [11] R. P. Stanley, Hilbert functions of graded algebras, *Adv. Math.* 28 (1978) 57–83.
- [12] Anonymous, Hilbert series of the invariants of tuples of 4×4 and 5×5 matrices, and a candidate series for plane septics, version 0.1.0-candidate, Evidence Press (2026), doi:10.5281/zenodo.22974308.